

■ Python Customer Churn Prediction

Insights • Recommendations • Business Impact

This report summarizes the machine learning results and key business actions derived from the Customer Churn Prediction project.

Dataset Quality

- **Insight:** 64,374 records, 12 features, no missing values or duplicates.
- **Recommendation:** Maintain strong data validation before every model training.
- **Business Impact:** High-quality data improves model reliability.

Logistic Regression

- **Insight:** Accuracy: 83.02%. A strong baseline model.
- **Recommendation:** Use as an interpretable benchmark.
- **Business Impact:** Simple and explainable predictions.

Decision Tree

- **Insight:** Highest accuracy: 99.83%.
- **Recommendation:** Validate with cross-validation to avoid overfitting.
- **Business Impact:** Excellent predictive performance.

Random Forest

- **Insight:** Accuracy: 99.79%. Robust and consistent.
- **Recommendation:** Suitable candidate for deployment.
- **Business Impact:** Handles complex patterns effectively.

Model Comparison

■ **Insight:** Decision Tree > Random Forest > Logistic Regression.

■ **Recommendation:** Compare precision, recall and F1-score before final selection.

■ **Business Impact:** Choose the model that generalizes best.

Feature Importance

■ **Insight:** Payment Delay, Support Calls and Tenure strongly influence churn.

■ **Recommendation:** Focus retention strategies on these drivers. Remove CustomerID in production models.

■ **Business Impact:** Better targeting can reduce churn.

Business Conclusion

■ **Insight:** ML successfully predicts customer churn and identifies key risk factors.

■ **Recommendation:** Use predictions for proactive customer retention.

■ **Business Impact:** Improves retention and long-term profitability.